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## Chapter 4

# GROUP THEORY

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### Introduction

A mathematical structure is a formal model which capture common properties or behavior of different sets. It consists of an abstract set, together with operations and relations which obey certain rules. In this chapter, we will discuss algebraic structure and develop a few basic properties of their structure.

### 4.1 Algebraic System

Algebraic system is a mathematical system consists of a set, with an operation on the set and the properties which are taken as the axioms of the system.

A binary operation on a set  $A$  is a function from  $A \times A$  to  $A$ . Addition and multiplication are some binary operations.

In general, the  $n$ -ary operation on  $A$  is a function from  $A^n$  to  $A$ .

A system consisting of a non-empty set  $S$  and one or more  $n$ -ary operations on the set  $S$  is the Algebraic System. It is denoted by  $\{S, f_1, f_2, f_3, \dots\}$  where  $f_1, f_2, f_3, \dots$  are operations on  $S$ .

A set  $S$  is said to be closed with respect to an operation if this operation on the members of  $S$  always produces another member of  $S$ . This property is called closure property.

In the set of all integers  $Z$  is closed with respect to the binary operation of addition and multiplication because addition and multiplication of two integers is also an integer. But subtraction is not a closure property in the set of all positive integers since subtraction of two positive integers need not be a positive integer.

### 4.1.1. Group

If  $G$  is a non-empty set and  $*$  is a binary operation such that  $(G, *)$  be an algebraic system, then  $(G, *)$  is called a group if the following conditions are satisfied,

1) **Closure Property**, If  $a, b \in G$ , then  $a * b \in G$ .

2) **Associative Property**, If  $a, b, c \in G$ , then  $a * (b * c) = (a * b) * c$

3) **Existence of Identity element**

There exists an identity element  $e \in G$  such that for any

$$a \in G, a * e = a = e * a.$$

4) **Existence of Inverse element**

For each  $a \in G$ , there exist  $a^{-1} \in G$  such that  $a * a^{-1} = e = a^{-1} * a$ .

### Abelian group

In a group  $(G, *)$  if  $a * b = b * a$  for all  $a, b \in G$ , then it is called a commutative or Abelian group.

Examples: The set  $Z$  of all integers with usual addition as operation is abelian group.

The set  $R$  of all real numbers with usual addition as operation is abelian group.

The set  $(R - \{0\})$  of all real numbers with usual multiplication as operation is abelian group.

$G$  consists of the real numbers  $\{-1, 1\}$  under multiplication of real numbers is a group.

### 4.1.2 Subgroup

Let  $G$  be a group,  $\phi \neq H$  is a subset of  $G$  then  $H$  is a subgroup of  $G$  if  $H$  itself is a group under the same binary operation of  $G$ .

For example, group  $(Z, +)$  is a subgroup of the group of all rational numbers with usual multiplication  $(Q, +)$ .

Group  $(E, +)$  of all even numbers with usual addition is a subgroup of the group of integers under addition.

### Order of an element of a group

The order of an element  $a$  of a group  $G$  is the smallest positive integer  $n$  such that  $a^n = e$ . It is denoted by  $o(a)$ . If no such integer exists, we say that  $G$  is infinite order.



Let  $\{1, -1, i, -i\}$  be a multiplication group with identity 1.  
 Then the order of the element 1 is 1 since  $1^1 = 1$ ,  
 order of the element  $-1$  is 2 since  $(-1)^2 = 1$   
 order of the element  $i$  is 4 since  $(i)^4 = 1$ ,  
 order of the element  $-i$  is 4 since  $(-i)^4 = 1$ .

### 4.1.3. Cyclic Group

A Group  $(G, *)$  is called a cyclic group if every element of  $G$  can be expressed as some power of a particular element  $a \in G$ . The element  $a$  is called the generator of the group  $G$  because for any  $x \in G, x = a^z$  for some  $z \in \mathbb{Z}$ .

For example, consider the group  $G = \{1, -1, i, -i\}$  then  $(G, \cdot)$  is a cyclic group generated by  $i$ , since  $(i)^1 = i, (i)^2 = -1, (i)^3 = -i, (i)^4 = 1$ .  
 $-i$  is also a generator of this group.

### 4.1.4. Homomorphism and isomorphism of groups

Let  $(X, \cdot)$  and  $(Y, *)$  are any two groups, the mapping  $f: (X, \cdot) \rightarrow (Y, *)$  is called group homomorphism if for any two elements

$$x_1, x_2 \in X, f(x_1 \cdot x_2) = f(x_1) * f(x_2).$$

If the function  $f$  is one - one and on to then it is a group isomorphism.  
 If there is a group isomorphism from  $X$  onto  $Y$  then the two semigroups  $(X, \cdot)$  and  $(Y, *)$  are said to be isomorphic.

### 4.1.5. Permutation group

A bijective mapping of a set into itself is a permutation.

Let  $S = \{a_1, a_2, a_3\}$  and  $p$  be a permutation on  $S$  ie,  $p$  is a bijective mapping from  $S \rightarrow S$

Suppose  $p(a_1) = a_2, p(a_2) = a_3, p(a_3) = a_1$ . This may be represented as  $\begin{pmatrix} a_1 & a_2 & a_3 \\ a_2 & a_3 & a_1 \end{pmatrix}$  or  $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$

If  $p_1$  and  $p_2$  are permutations,  $p_1 \circ p_2$  is a permutation called composition of functions.

Suppose  $p_1 = \begin{pmatrix} a_1 & a_2 & a_3 \\ a_2 & a_3 & a_1 \end{pmatrix}$  and  $p_2 = \begin{pmatrix} a_1 & a_2 & a_3 \\ a_3 & a_2 & a_1 \end{pmatrix}$

$$\text{Then } p_1 \circ p_2 = \begin{pmatrix} a_1 & a_2 & a_3 \\ a_1 & a_3 & a_2 \end{pmatrix}$$

$$\text{i.e. } p_1 \circ p_2(a_1) = p_1(p_2(a_1)) = p_1(a_3) = a_1$$

$$p_1 \circ p_2(a_2) = p_1(p_2(a_2)) = p_1(a_2) = a_3$$

$$p_1 \circ p_2(a_3) = p_1(p_2(a_3)) = p_1(a_1) = a_2$$

The set  $G$  of all permutations on a nonempty set  $X$  under the binary operation 'o' of composition of permutations, is a group called permutation group.

Permutation group is a group made up of elements which are permutations of a set.

If  $X = \{1, 2, 3, \dots, n\}$ , the permutation group is also called symmetric group denoted by  $S_n$ . The number of elements of  $S_n$  is  $n!$ .

For example, let  $S_3$  be the set all permutations on the set  $S = \{1, 2, 3\}$  is a group under the operation of composition of permutations.

### Addition Modulo $n$

If  $a$  and  $b$  are two integers, then addition modulo  $n$ ,  $\oplus_n$ , is the remainder  $t$  when  $a + b$  is divided by  $n$  which is denoted by

$$a \oplus_n b \equiv t$$

For example  $3 + 4 \equiv 2 \pmod{5}$

$(Z_n, \oplus_n)$ ,  $n \in \mathbb{Z}^+$ ,  $n > 1$ , is an abelian group of integers under addition modulo  $n$  where  $Z_n$  is the set of all integers when divided by  $n$

$$Z_n = \{0, 1, 2, \dots, n-1\}.$$

The generators of  $Z_n$  are the elements of  $\mathbb{Z}$  that are relatively prime to  $n$ . The generators of  $Z_{12}$  are 1, 5, 7, 11.

$(Z_4, \oplus_4)$  is an abelian group where  $Z_4 = \{0, 1, 2, 3\}$ .

### Theorem 4.1.1

For every group  $G$  prove that

a) The identity element of  $G$  is unique b) The inverse of each element of  $G$  is unique.

**Proof:**

a) Assume that  $e_1, e_2$  be the two identity elements in the group  $G$ .

Since  $e_1$  is the identity element in  $G$ , and  $e_2 \in G$ ,  $e_1 e_2 = e_2 = e_2 e_1$  (1)

Since  $e_2$  is the identity element in  $G$ , and  $e_1 \in G$ ,  $e_1 e_2 = e_1 = e_2 e_1$  (2)



From (1) and (2)  $e_1 = e_2$ . Hence the identity element in a group is unique.

a) Let  $a \in G$ , and suppose  $b$  and  $c$  are inverse elements of  $a$ . Since  $b$  is the inverse of  $a$ ,  $ab = e = ba$ . Since  $c$  is the inverse of  $a$ ,  $ac = e = ca$  where  $e$  is the identity element of  $G$ .

Then,  $b = eb = (ca)b = c(ab) = ce = c$ .

Hence the inverse element of  $G$  is unique.

### Theorem 4.1.2

Let  $H$  is a nonempty subset of  $G$ . Then  $H$  is subgroup of  $G$  if and only if a) for all  $a, b \in H \Rightarrow ab \in H$       b)  $a \in H \Rightarrow a^{-1} \in H$ .

**Proof:**

Assume that  $H$  is a subgroup of  $G$ . Then  $H$  is a group under the same binary operation in  $G$ . Hence  $H$  satisfies all conditions of a group.

Therefore,  $a, b \in H \Rightarrow ab \in H$  (Closure property)

$a \in H \Rightarrow a^{-1} \in H$  (Existence of inverse)

Conversely, assume that  $\phi \neq H \subseteq G$  and  $H$  satisfying conditions  $a, b \in H \Rightarrow ab \in H$ ,  $a \in H \Rightarrow a^{-1} \in H$ .

Prove that  $H$  is subgroup of  $G$ .

For all  $a, b, c \in H \Rightarrow (ab)c = a(bc)$  in  $G$  so  $(ab)c = a(bc)$  in  $H$ .

Associative property is true in  $H$ .

If  $a \in H \Rightarrow a^{-1} \in H$ , by closure property,  $aa^{-1} \in H \Rightarrow e \in H$  identity element exists in  $H$ .  $H$  satisfies all the properties of a group. Therefore,  $H$  is a subgroup of  $G$ .

### Theorem 4.1.3

If  $G$  is a group and  $\phi \neq H \subseteq G$  with  $H$  is finite, then  $H$  is a subgroup of  $G$  if and only if  $H$  is closed under binary operation of  $G$ .

**Proof:**

Assume that  $\phi \neq H \subseteq G$  and  $H$  is a finite subset of  $G$ . Prove that  $H$  is closed under binary operation of  $G$ . Since  $H \subseteq G$  and  $H$  is a subset of  $G$  then  $H$  itself is a group under the same operation on  $G$ . Therefore,  $H$  is closed under the binary operation of  $G$ .

Conversely, assume that  $H$  is a finite subset of  $G$  and is closed under binary operation of  $G$ . i.e.  $a, b \in H \Rightarrow ab \in H$  for all  $a, b \in H$ . To prove that  $H$  is a subgroup of  $G$ .

Let  $a \in H$ ,  $a \in H \Rightarrow a.a \in H \Rightarrow a^2 \in H$ ,  $a^3 \in H$ ,  $a^4 \in H, \dots, a^n \in H, \dots$

(by closure property). Since  $H$  is finite there must be repetitions in the collection.

That is, for some  $r > s > 0$ ,  $a^r = a^s$ . By cancellation in  $G$ ,

$a^{r-s} \in e \in H$  is the identity element in  $H$ . Hence, identity element exists in  $H$ .

Since  $r - s - 1 \geq 0$ ,  $a^{r-s-1} \in H \Rightarrow a^{r-s} a^{-1} \in H$ ,  $e \cdot a^{-1} \in H \Rightarrow a^{-1} \in H$ .

Inverses exist in  $H$ . Therefore,  $H$  is a subgroup of  $G$ .

#### Theorem 4.1.4

Let  $(G, \cdot), (H, *)$  be groups with respective identities  $e_G, e_H$ . If  $f: G \rightarrow H$  is a homomorphism, then

- a)  $f(e_G) = e_H$                       b)  $f(a^{-1}) = [f(a)]^{-1}$  for all  $a \in G$
- c)  $f(a^n) = [f(a)]^n$  for all  $a \in G$  and  $n \in \mathbb{Z}$ .
- d)  $f(S)$  is a subgroup of  $H$  for each subgroup  $S$  of  $G$ .

**Proof:**

- a)  $e_H * f(e_G) = f(e_G)$                       since  $e_H$  is the identity element in  $H$ .  
 $= f(e_G \cdot e_G)$                       since  $e_G$  is the identity element in  $G$ .  
 $= f(e_G) * f(e_G)$                       since  $f$  is a homomorphism.

Therefore, by right cancellation,  $e_H = f(e_G)$ .

- b) Let  $a, a^{-1} \in G$ ,  $a \cdot a^{-1} = e_G$  is the identity in  $G$ .

$f(a \cdot a^{-1}) = f(e_G) = e_H$  is the identity in  $H$ .

Since  $f$  is a homomorphism  $f(a \cdot a^{-1}) = f(a) * f(a^{-1}) = e_H$ .

i.e. the inverse of  $f(a)$  is  $f(a^{-1})$  or  $[f(a)]^{-1} = f(a^{-1})$  for all  $a \in G$

- c) For all  $a \in G$ , by closure property,  $a \cdot a \in G$ ,  $a \cdot a \cdot a \in G \dots$ ,  
 $a^n = a \cdot a \cdot \dots \cdot a \in G$

Since  $f$  is a homomorphism,

$$f(a \cdot a \cdot \dots \cdot a \text{ } n \text{ times}) = f(a) * f(a) * \dots \cdot n \text{ times},$$

i.e.  $f(a^n) = [f(a)]^n$ .

- d) If  $S$  is a subgroup of  $G$ ,  $S \neq \emptyset$ ,  $f(S) \neq \emptyset$ .

Let  $x, y \in f(S)$ . Then  $x = f(a)$  and  $y = f(b)$ , for  $a, b \in S$ . Since  $S$  is a subgroup of  $G$ ,  $x * y = f(a) * f(b) = f(a \cdot b) \in f(S)$ .

Finally,  $x^{-1} = [f(a)]^{-1} = f(a^{-1}) \in f(S)$  since  $a^{-1} \in S$  by theorem 4.1.2  
 $f(S)$  is a subgroup of  $H$ .

**Theorem 4.1.5**

Let  $a \in G$  with  $o(a) = n$ . If  $k \in \mathbb{Z}$  and  $a^k = e$  then  $n$  divides  $k$ .

**Proof:**

Since  $o(a) = n$ ,  $n$  is the least positive integer such that  $a^n = e$ .

By division algorithm,  $k = qn + r$ ,  $0 \leq r < n$ ,  $r$  is a positive integer.

Then  $e = a^k = a^{qn+r} = (a^n)^q a^r = e^q a^r = a^r$ , which is a contradiction since  $r < n$ ,  $n$  is the least positive integer such that  $a^n = e$ . The only possibility is  $r = 0$ . Hence  $k = qn$ . Thus  $n$  divides  $k$ .

**Theorem 4.1.6**

Prove that every subgroup of a cyclic group is cyclic.

**Proof:**

Let  $G$  be a cyclic group and  $a$  be the generator of  $G$ . Let  $H$  be the subgroup of  $G$ . Then each element of  $H$  has the form  $a^k$  for some  $k \in \mathbb{Z}$ .

For  $H \neq \{e\}$ , let  $t$  be the smallest positive integer such that  $a^t \in H$ .

Let  $b \in H \Rightarrow b = a^s$  for some  $s \in \mathbb{Z}$ . By division algorithm,  $s = qt + r$  where  $q, r \in \mathbb{Z}$ ,  $0 \leq r < t$ .  $a^s = a^{qt+r} = a^{qt} a^r$ .

Consequently,  $a^r = a^s \cdot a^{-qt} = b \cdot (a^t)^{-q}$ . Since  $H$  is a subgroup of  $G$ ,  $a^t \in H \Rightarrow (a^t)^{-q} \in H$ ,

That is,  $(a^t)^{-q} \cdot b \in H \Rightarrow (a^t)^{-q} a^s \in H \Rightarrow a^{s-qt} \in H \Rightarrow a^r \in H$  which is a contradiction since  $0 \leq r < t$  and  $a^t$  is the smallest integer such that  $a^t \in H$ . Therefore the only possibility is  $r = 0$ .

Thus, we have  $s = qt \Rightarrow a^s = a^{qt} = (a^t)^q$ . So, every element in  $H$  is generated by  $a^t$ . Thus  $H$  is cyclic.

**Theorem 4.1.7**

Every finite group of order  $n$  is isomorphic to a permutation group of order  $n$ .

**Proof**

Let  $G$  be a given group of order  $n$ . For each element  $a$  in  $G$ , we define a mapping  $f_a: G \rightarrow G$  by  $f_a(x) = ax$  for all  $x$  in  $G$ .



Now  $f_a$  is one-to-one since  $f_a(x) = f_a(y) \Rightarrow ax = ay \Rightarrow x = y$

Let  $b$  be an arbitrary element in  $G$ . Then  $x = a^{-1}b$  is in  $G$ , and for this particular  $x$  we have  $f_a(x) = ax = a(a^{-1}b) = b$ . Therefore,  $f_a$  is on-to. Thus  $f_a$  is a permutation on the set of elements of  $G$ .

We shall show that the set  $G' = \{f_a/a \in G\}$  forms a group of permutations. Since mapping composition is always associative, we need only show that  $G'$  is closed, has an identity, and contains inverses.

For any  $f_a$  and  $f_b$  in  $G'$ , we have

$$f_a f_b(x) = f_a(f_b(x)) = f_a(bx) = a(bx) = (ab)(x) = f_{ab}(x)$$

Therefore  $G'$  is closed.

Consider  $f_a \cdot f_e$  where  $e \in G$ .

$$f_a \cdot f_e(x) = f_a(x) \Rightarrow f_a(ex) = f_a(x)$$

$$\text{i.e. } f_a \cdot f_e = f_a = f_e \cdot f_a$$

$\Rightarrow f_e$  is the identity of  $G'$

$$\text{We have } f_a f_{a^{-1}} = f_{aa^{-1}} = f_e$$

$$f(a)^{-1} = f_{a^{-1}} \text{ is in } G' \therefore \text{inverse exist in } G'$$

Consequently,  $G'$  is a group of permutations.

To show that  $G$  is isomorphic to  $G'$ .

The mapping  $\varphi: G \rightarrow G'$  defined by  $\varphi(a) = f_a$  is a bijective function.

and  $\varphi(a)\varphi(b) = f_a f_b = f_{ab} = \varphi(ab) \therefore \varphi$  is an isomorphism.

### **Theorem 4.1.8**

In any group  $(G, *)$  the power of any fixed element  $a \in G$  constitute a subgroup of  $G$ .

#### **Proof**

Let  $G' = \{a, a^2, a^3 \dots\}$  consists of all powers of an element  $a$  of  $G$ .

To prove that  $G'$  is a subgroup of  $G$

Closure property and associative property are true because

$$a^r \cdot a^s = a^{r+s} \in G' \text{ and } a^r \cdot (a^s \cdot a^t) = a^r (a^{s+t}) = a^{r+(s+t)} = a^{(r+s)+t}$$



$$= (a^r \cdot a^s) a^t$$

$a^m \cdot a^0 = a^{m+0} = a^m = a^0 \cdot a^m \Rightarrow a^0 = e$  is the identity element.

$a^r \cdot a^{-r} = a^0 = e = a^{-r} \cdot a^r$  Inverse element exists.

It follows that  $G'$  is a subgroup of  $G$ .

### Worked Example. 4.1

#### Example 4.1.1

Show that  $(A, *)$  be an abelian group where  $A = \{a \in \mathbb{Q} | a \neq -1\}$  and for any  $a, b \in A$ ,  $a * b = a + b + ab$ .

**Solution:**

1. For any  $a, b \in A$ ,  $a * b = a + b + ab \in A \quad \therefore$  Closure property is satisfied

2. For any  $a, b, c \in A$ ,  $(a * b) * c = (a + b + ab) * c$   

$$= a + b + ab + c + (a + b + ab)c$$

$$= a + b + c + ab + ac + bc + abc$$

$$a * (b * c) = a * (b + c + bc) = a + b + bc + c + (b + c + bc)a$$

$$= a + b + c + ab + ac + bc + abc = (a * b) * c$$

Associative property is true

3. For all  $a \in A$  there exist  $e \in A$  such that  $a * e = a$ .

So  $a * e = a + e + ae = a \Rightarrow e(1 + a) = 0$ . Since  $a$  is arbitrary and  $a \neq -1$ ,  $1 + a \neq 0$ . Therefore  $e = 0$  is the identity element. So, identity element exists.

4. For any  $a \in A$  there exist  $b \in A$  such that  $a * b = 0$ . ie.  $a + b + ab = 0$   
 . Thus  $b = \frac{-a}{1+a}$  is the inverse of  $a$ . So inverse exist.

5.  $a * b = a + b + ab = b + a + ba = b * a$ . So  $*$  is commutative.  $\therefore (A, *)$  is an abelian group.

#### Example 4.1.2

Show that  $Q^+$  of all positive rational numbers form an abelian group under the operation  $*$  defined by  $a * b = \frac{1}{2} ab$ ,  $a, b \in Q^+$ .

**Solution:**

1. For any  $a, b \in Q^+$ ,  $a * b = \frac{1}{2} ab \in Q^+$ , Closure property is satisfied.

2. For any  $a, b, c \in A$ ,  $(a * b) * c = \left(\frac{1}{2} ab\right) * c = \frac{1}{4} (ab)c = \frac{1}{4} a(bc)$   

$$= a * (b * c)$$

Associative property is true

3. For all  $a \in A$  there exist  $e \in A$  such that  $a * e = \frac{1}{2} ae = a$

So  $e = 2 \in Q^+$  is the identity element.

4. For any  $a \in A$  there exist  $b \in A$  such that  $a * b = 2$ . ie.  $\frac{1}{2} ab = 2$ .

Thus  $b = \frac{4}{a} \in Q^+$  is the inverse element of  $a$ . So inverse exist.

5.  $a * b = \frac{1}{2} ab = \frac{1}{2} ba = b * a$ . So  $*$  is commutative.  $\therefore (Q^+, *)$  is an abelian group.

### Example 4.1.3

Show that  $(A, *)$  be a non abelian group where  $A = R \times R$  and for any  $a, b \in A$ ,  $(a, b) \cdot (c, d) = (ac, bc + d)$

**Solution:**

1.  $(a, b) \cdot (c, d) = (ac, bc + d) \in A$  Closure property is satisfied

2.  $[(a, b) \cdot (c, d)] \cdot (e, f) = (ac, bc + d) \cdot (e, f)$   
 $= (ace, (bc + d)e + f)$   
 $= (ace, bce + de + f)$

$(a, b) \cdot [(c, d) \cdot (e, f)] = (a, b) \cdot (ce, de + f) = (ace, bce + de + f)$   
 $= [(a, b) \cdot (c, d)] \cdot (e, f).$

Associative property is true

3. For all  $(a, b) \in A$  there exist  $(e_1, e_2) \in A$  such that

$$(a, b) \cdot (e_1, e_2) = (a, b)$$

$$(a, b) \cdot (e_1, e_2) = (ae_1, be_1 + e_2) = (a, b)$$

Equating,  $ae_1 = a$ ,  $be_1 + e_2 = b$ . ie,  $e_1 = 1$  and  $e_2 = 0$ .

Thus  $(1, 0) \in A$  is the identity element. So, identity exist in  $A$ .

4. For any  $(a, b) \in A$  there exist  $(c, d) \in A$  such that  $(a, b) \cdot (c, d) = (1, 0)$

$$\Rightarrow (ac, bc + d) = (1, 0).$$

Equating,  $ac = 1$ ,  $bc + d = 0$  ie,  $c = a^{-1}$  and  $d = -ba^{-1}$

$\therefore$  the inverse of  $(a, b)$  is  $(a^{-1}, -ba^{-1}) \in A$ . Inverse exists.

5.  $(a, b) \cdot (c, d) = (ac, bc + d) \neq (ca, ad + b) = (c, d) \cdot (a, b)$

$\therefore (A, *)$  is a non abelian group.

**Example 4.1.4**

Show that any group  $G$  is abelian if and only if  $(ab)^2 = a^2 b^2$  for all  $a, b \in G$ .

**Solution:**

Suppose  $G$  is abelian

$$\begin{aligned} \text{Now } (ab)^2 &= (ab)(ab) \\ &= a(ba)b \quad \text{by associative property} \\ &= a(ab)b \quad \text{since } G \text{ is abelian} \\ &= (aa)(bb) \quad \text{by associative property} \\ &= a^2 b^2 \end{aligned}$$

$$\begin{aligned} \text{Suppose } (ab)^2 &= a^2 b^2 \\ (ab)(ab) &= (aa)(bb) \\ a(ba)b &= a(ab)b \quad \text{by associative property} \\ (ba)b &= (ab)b \quad \text{by cancellation property} \\ (ba) &= (ab) \quad \text{by cancellation property} \end{aligned}$$

Therefore,  $G$  is abelian.

**Example 4.1.5**

Prove that commutative property is invariant under group homomorphism.

**Solution:**

Let  $f: A \rightarrow B$  be a group homomorphism.

Suppose  $A$  is abelian

Then for any  $a_1, a_2 \in A$  there exist  $b_1, b_2 \in B$  such that  $f(a_1) = b_1$  and  $f(a_2) = b_2$ .

$$\begin{aligned} \text{Now } b_1 b_2 &= f(a_1) f(a_2) \\ &= f(a_1 a_2) \quad \text{since } f \text{ is a homomorphism} \\ &= f(a_2 a_1) \quad \text{since } G \text{ is abelian} \\ &= f(a_2) f(a_1) \quad \text{since } f \text{ is a homomorphism} \\ &= b_2 b_1. \end{aligned}$$

$\therefore B$  is commutative.

**Example 4.1.6**

If  $G$  is a group, prove that for all a)  $(a^{-1})^{-1} = a$ , b)  $(ab)^{-1} = b^{-1}a^{-1}$ .

**Solution:**

For any  $a \in G$ ,  $aa^{-1} = e$ , the identity element in  $G$ . Which means the inverse of  $a^{-1}$  is  $a$

ie,  $(a^{-1})^{-1} = a$ .



$$\begin{aligned}
 \text{Consider } (ab)(b^{-1}a^{-1}) &= a(bb^{-1})a^{-1} && \text{by associative property} \\
 &= (ae)a^{-1} && \text{Inverse exist in } G \\
 &= aa^{-1} && \text{Identity exists in } G \\
 &= e && \text{Inverse exists in } G
 \end{aligned}$$

Therefore, the inverse of  $ab$  is  $b^{-1}a^{-1}$ . So  $(ab)^{-1} = b^{-1}a^{-1}$ .

#### Example 4.1.7

Prove that if  $H$  and  $K$  are subgroup of  $G$  then  $H \cap K$  is a subgroup of  $G$ .

##### Solution:

Since  $H$  and  $K$  are subgroups of  $G$ ,  $e \in H, e \in K \Rightarrow e \in H \cap K$ . The identity element exists.

Let  $x, y \in H \cap K \Rightarrow x, y \in H$  and  $x, y \in K$

$\Rightarrow xy \in H$  and  $xy \in K$  by closure property of  $H$  and  $K$ .

$\Rightarrow xy \in H \cap K$

Closure property is satisfied.

Let  $a \in H \cap K \Rightarrow a \in H$  and  $a \in K$

$\Rightarrow a^{-1} \in H$  and  $a^{-1} \in K$  since  $H$  and  $K$  are subgroups

$\Rightarrow a^{-1} \in H \cap K$

Therefore, the inverse element exists in  $H \cap K$ .  $\therefore H \cap K$  is a subgroup of  $G$ .

#### Example 4.1.8

Prove that every cyclic group is abelian.

##### Solution:

**Let**  $(G, *)$  be a cyclic group generated by  $a$ . i.e.  $G = \{a^n \mid n \in \mathbb{Z}\}$ .

Let  $x, y \in G$ ,  $x = a^n$ ,  $y = a^m$ , where  $m, n \in \mathbb{Z}$ .

$$x * y = a^n \cdot a^m = a^{n+m} = a^{m+n} = a^m \cdot a^n = y * x.$$

$\therefore G$  is abelian.

#### Example 4.1.9

In the group  $S_4$ , let  $\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix}$  and  $\beta = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix}$

Determine  $\alpha \circ \beta$ ,  $\beta \circ \alpha$ ,  $\alpha^2$

##### Solution:

$$\alpha \circ \beta = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix}$$

$$\beta \circ \alpha = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 2 & 1 & 4 \end{pmatrix}$$

$$\alpha^2 = \alpha \circ \alpha = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix}$$

### Example 4.1.10

If  $X = \{1, 2, 3\}$  List all permutations  $S_3$  of  $X$  and prove that  $(S_3, \circ)$  is a group where the operation  $\circ$  is the composition of permutations. Is it abelian?

#### Solution:

Consider  $(S_3, \circ)$  where  $X = \{1, 2, 3\}$  is a group under the operation of composition of permutation.

$$S_3 = (f_1, f_2, f_3, f_4, f_5, f_6) \text{ where } f_1 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}, f_2 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$$

$$f_3 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} \quad f_4 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix} \quad f_5 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} \quad f_6 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}$$

$$f_2 \circ f_4 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} \circ \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} = f_6$$

$$f_3 \circ f_4 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} \circ \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} = f_5$$

| $\circ$ | $f_1$ | $f_2$ | $f_3$ | $f_4$ | $f_5$ | $f_6$ |
|---------|-------|-------|-------|-------|-------|-------|
| $f_1$   | $f_1$ | $f_2$ | $f_3$ | $f_4$ | $f_5$ | $f_6$ |
| $f_2$   | $f_2$ | $f_3$ | $f_1$ | $f_6$ | $f_4$ | $f_5$ |
| $f_3$   | $f_3$ | $f_1$ | $f_2$ | $f_5$ | $f_6$ | $f_4$ |
| $f_4$   | $f_4$ | $f_5$ | $f_6$ | $f_1$ | $f_2$ | $f_3$ |
| $f_5$   | $f_5$ | $f_6$ | $f_4$ | $f_3$ | $f_1$ | $f_2$ |
| $f_6$   | $f_6$ | $f_4$ | $f_5$ | $f_2$ | $f_3$ | $f_1$ |

From the table closure property, Associative property are true. For example

$$(f_3 \circ f_4) \circ f_5 = f_5 \circ f_5 = f_1 = f_3 \circ (f_4 \circ f_5)$$

$$f_1 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix} \text{ is the identity element.}$$

$$\text{Also } f_1^{-1} = f_1, f_2^{-1} = f_3, f_3^{-1} = f_2, f_4^{-1} = f_4, f_5^{-1} = f_5, f_6^{-1} = f_6.$$

Thus the inverse element for each element exist. So  $(S_3, \circ)$  is a group.

The symmetric group  $(S_3, \circ)$  is not abelian since

$$f_3 \circ f_4 = f_5 \neq f_4 \circ f_3 = f_6.$$

**Example 4.1.11**

Prove that  $(Z_6, \oplus_6)$  is an abelian group.

**Solution:**

| $\oplus_6$ | 0 | 1 | 2 | 3 | 4 | 5 |
|------------|---|---|---|---|---|---|
| 0          | 0 | 1 | 2 | 3 | 4 | 5 |
| 1          | 1 | 2 | 3 | 4 | 5 | 0 |
| 2          | 2 | 3 | 4 | 5 | 0 | 1 |
| 3          | 3 | 4 | 5 | 0 | 1 | 2 |
| 4          | 4 | 5 | 0 | 1 | 2 | 3 |
| 5          | 5 | 0 | 1 | 2 | 3 | 4 |

Closure property and associative property is true. 0 is the identity element.

Inverse of 0 is 0, Inverse of 1 is 5, Inverse of 2 is 4, Inverse of 3 is 3, Inverse of 4 is 2, Inverse of 5 is 1. Commutative property is true.

$\therefore (Z_6, \oplus_6)$  is an abelian group.

**Example 4.1.12**

$Z_n$  is the group of integers mod  $n$  under modular addition. List the generators of

(i)  $Z_{12}$  (ii)  $Z_p$ , where  $p$  is prime.

**Solution:**

$$Z_{12} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$$

(i) Generator of  $Z_{12} = \{1, 5, 7, 11\}$  since  $\gcd\{1, 12\} = 1$ ,  $\gcd\{5, 12\} = 1$ ,  $\gcd\{7, 12\} = 1$ ,  $\gcd\{11, 12\} = 1$

(ii) If  $p$  is prime,  $Z_p = \{1, 2, 3, 4, \dots, p-1\}$

**Example 4.1.13**

Let  $G = (Z_6, \oplus_6)$ . If  $H = \{0, 2, 4\}$  Prove that  $H$  is a subgroup of  $G$ .

**Solution:**

$Z_6 = \{0, 1, 2, 3, 4, 5\}$   $H$  is subset of  $G$ .

| $\oplus_6$ | 0 | 2 | 4 |
|------------|---|---|---|
| 0          | 0 | 2 | 4 |
| 2          | 2 | 4 | 0 |
| 4          | 4 | 0 | 2 |



$H$  is a subgroup of  $G$  under the operation  $\oplus_6$ . 0 is the identity element.

**Example 4.1.14**

Let  $G = (Z, +)$  and  $H = (Z_4, \oplus_4)$  are groups.  $f: G \rightarrow Z_4$  be a function from  $G$  to  $H$  defined by  $f(x) = [x] = \{x + 4k | k \in Z\}$ . Prove that  $f$  is a group homomorphism.

**Solution:**

For all  $x, y \in G$   $f(x + y) = [x + y] = [x] + [y] = f(x) + f(y)$ . Since  $G$  and  $H$  are groups and  $f$  preserves homomorphism,  $f$  is a group homomorphism.

**Example 4.1.15**

Prove that group  $H = (Z_4, \oplus_4)$  is cyclic.

**Solution:**

Let  $Z_4 = \{0, 1, 2, 3\}$

| $\oplus_4$ | 0 | 1 | 2 | 3 |
|------------|---|---|---|---|
| 0          | 1 | 1 | 2 | 3 |
| 1          | 1 | 2 | 3 | 0 |
| 2          | 2 | 3 | 0 | 1 |
| 3          | 3 | 0 | 1 | 2 |

The group is cyclic since every element of the group can be expressed as the power of the generator.

The generators of  $Z_4$  are  $[1]$  and  $[3]$  since

$$1^0 = 1, \quad 1^1 = 1, \quad 1^2 = 1 \oplus_4 1 = 2, \quad 1^3 = 1 \oplus_4 1 \oplus_4 1 = 3$$

$$3^0 = 1, \quad 3^1 = 3, \quad 3^2 = 3 \oplus_4 3 = 2, \quad 3^3 = 3 \oplus_4 3 \oplus_4 3 = 1$$

Every element in the group is the power of 1 and 3.

Therefore  $H$  is a cyclic group.

**Example 4.1.16**

Let  $f: G \rightarrow H$  be a group homomorphism. If  $G$  is abelian prove that  $H$  is abelian.

**Solution:**

For  $x, y \in G$ ,  $f(x) = a \in H$ ,  $f(y) = b \in H$ .

Now,  $a * b = f(x) * f(y) = f(xy)$  since  $f$  is a homomorphism.

$$\begin{aligned}
 &= f(yx) \text{ since } G \text{ is abelian} \\
 &= f(y) * f(x) \text{ since } f \text{ is a homomorphism.} \\
 &= b * a
 \end{aligned}$$

Therefore,  $H$  is abelian.

## 4.2. Cosets

If  $H$  is a subgroup of  $G$ , then for each  $a \in G$  the set  $aH = \{ah: h \in H\}$  is called the left coset of  $H$  in  $G$ . The set  $Ha = \{ha: h \in H\}$  is the right cosets of  $H$  in  $G$ .

If the operation in  $G$  is addition, then  $aH$  is  $a + H$ . Then the cosets are

$$a + H = \{a + h: h \in H\} \quad \text{and} \quad H + a = \{h + a: h \in H\}$$

For example, let  $G = (\mathbb{Z}, +)$  be the group of integers under addition. Let  $H = \{\dots, -8, -4, 0, 4, 8, \dots\}$ . Left cosets of  $H$  in  $G$  is

$$0 + H = H$$

$$1 + H = \{\dots - 7, -3, 1, 5, 9, \dots\}$$

$$2 + H = \{\dots - 6, -2, 2, 6, 10, \dots\}$$

$$3 + H = \{\dots - 5, -1, 3, 7, 11, 15, \dots\}$$

### Theorem: 4.2.1

If  $H$  is a subgroup of a finite group  $G$ , then for all  $a, b \in G$

$$a) |aH| = |H| \quad b) \text{ Either } aH = bH \text{ or } aH \cap bH = \emptyset.$$

**Proof:**

a) Since  $aH = \{ah: h \in H\}$  then  $|aH| \leq |H|$ . If  $|aH| < |H|$  we have  $ah_i = ah_j$  where  $h_i$  and  $h_j$  are distinct elements of  $H$ . By left cancellation  $h_i = h_j$ . So  $|aH| = |H|$ .

b) If  $aH \cap bH \neq \emptyset$ . Let  $c \in aH \cap bH$  then  $c \in aH$  and  $c \in bH$ .

Let  $c = ah_1 = bh_2$ , for some  $h_1, h_2 \in H$ . Then  $a = h_1^{-1} b h_2$  and  $b = h_2^{-1} a h_1$

If  $x \in aH$  then  $x = ah$  for some  $h \in H$ .

Hence,  $x = ah = (bh_2h_1^{-1})h = b(h_2h_1^{-1}h) \in bH$ . Then  $aH \subseteq bH$

If  $y \in bH$  then  $y = bh$  for some  $h \in H$ . So  $y = bh = (h_2^{-1}ah_1)h = (ah_2^{-1}h_1)h \in aH$ .  
 Then  $aH \subseteq bH$ . Therefore  $aH$  and  $bH$  are either identical or disjoint.

### 4.2.2. Lagrange's Theorem

#### Statement

If  $G$  is a finite group of order  $n$  with  $H$  is a subgroup of  $G$  of order  $m$ , then  $m$  divides  $n$ .

#### Proof

If  $H = G$  the result follows. If  $m < n$  there exist an element  $a \in G - H$ .

Since  $a \notin H$ ,  $aH \neq H$  so that  $aH \cap H = \emptyset$ .

If  $G = aH \cup H$ ,  $|G| = |aH| + |H|$ ,  $n = m + m = 2m$ . Then  $m$  divides  $n$  and the theorem follows.

$G \neq aH \cup H$  there exists an element  $b \in G - \{aH \cup H\}$  with

$$bH \cap H = \emptyset = bH \cap aH$$

$|G| = |aH| + |bH| + |H| = m + m + m$ . That is  $n = 3m$ . Then  $m$  divides  $n$  and the theorem follows.

Otherwise there exist  $c \in G - \{aH \cup bH \cup H\}$

Since the group is finite, this process terminates and

$$G = a_1H \cup a_2H \cup \dots \cup a_kH.$$

So  $n = m + m + m + \dots k \text{ times} = km$ . Therefore  $m$  divides  $n$ . Thus, the theorem follows in all cases.

### Deductions from Lagrange's Theorem

a) The order of any element of a finite group is a divisor of the order of the group.

#### Proof

Let  $G$  be the finite group and  $a \in G$ . Let the order of  $a$  is  $m$ .

$$\text{Then } a^m = e$$

Let  $H$  be a cyclic subgroup of  $G$  generated by  $a$ .



Then  $H = \{a, a^2, a^3, \dots, a^m = e\} \therefore O(H) = m$ .

By Lagrange's Theorem,  $O(H)$  is a divisor of  $O(G)$ . ie,  $m$  is a divisor of  $O(G)$  ie,  $o(a)$  is a divisor of  $O(G)$

**b) If  $G$  is a finite group of order  $n$ , then  $a^n = e$  for any  $a \in G$ .**

If  $m$  is the order of  $a$ , then  $a^m = e$ . Then  $m$  is a divisor of  $n$ . ie,  $n = km$ .

Now,  $a^n = a^{km} = (a^m)^k = e^k = e$ .

**c) Every group of prime order is cyclic.**

Let  $G$  be a group with  $o(G) = p$  where  $p$  is prime. Let  $a \neq e \in G$ .  $H$  is a cyclic sub group of  $G$  generated by  $a$ . By Lagrange's theorem  $O(H) \mid p$ . So  $O(H) = 1$  or  $p$  since  $p$  is prime.

If  $O(H) = 1$ , then  $a = e$  which is not possible since  $a \neq e$ .

Hence  $o(H) = p$

Therefore  $G = H$  and  $H$  is cyclic. Hence  $G$  is cyclic.

### Exercise 4.1 & 4.2

1. Prove that the set  $\{-1, 1\}$  is a group under the operation multiplication.

2. Prove that  $(\mathbb{Z}, *)$  is an abelian group under the operation  $x * y = x + y + 1$

3. Let  $S = R^* \times R$  where  $R$  is the set of all real numbers and  $R^*$  is the set of all nonzero real numbers. Define a binary operation  $\bullet$  on  $S$  by  $(u, v) \bullet (x, y) = (ux, vx + y)$ . Prove that  $(S, \bullet)$  is a non abelian group.

4. If  $f: G \rightarrow H$  be a group homomorphism onto  $H$ . If  $G$  is abelian prove that  $H$  is abelian.

5. If  $H$  and  $K$  are subgroup of a group  $G$ , where  $e$  is the identity of  $G$ . Prove that if  $|H| = 10$  and

$|K| = 21$ , then  $H \cap K = \{e\}$ .

6. Let  $G = \{1, a, a^2, a^3\} (a^4 = 1)$  be a group and  $H = \{1, a^2\}$  is a subgroup of  $G$  under multiplication. Find all right cosets of  $H$  in  $G$ .

7. In the group  $S_6$ , let  $\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 4 & 5 \end{pmatrix}$  and

$\beta = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 1 & 5 & 3 & 4 \end{pmatrix}$

Determine  $\alpha \circ \beta$ ,  $\beta \circ \alpha$ ,  $\alpha^2$

### Answers:

1. Closure property and associative property are true. 1 is the identity and the inverse of -1 is -1.

2. Closure property and associative property are true.  $e = -1$  is the identity,  $-a - 2$  is the inverse of  $a$ .

3. Closure property and associative property are true.  $(1,0)$  is the identity and  $(u^{-1}, -vu^{-1})$  is the inverse.  $(u,v) \cdot (x,y) \neq (x,y) \cdot (u,v)$ .

4.  $f(a).f(b) = f(ab) = f(ba) = f(b).f(a)$

5. If  $x \in H \cap K$ . If the order of  $x$  is  $n$ , then  $n$  divides both 10 and 21. Since  $\gcd(10,21)=1$ ,  $n=1$  So  $x = e, H \cap K = \{e\}$

6.  $H1 = H$ ,  $Ha = \{a, a^2\}$ ,  $Ha^2 = \{a^2, a^4\} = \{a^2, 1\} = H$ ,  $Ha^3 = \{a^3, a^5\} = \{a^3, a\} = Ha$

7.  $\alpha \circ \beta = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 5 & 2 & 3 & 4 \end{pmatrix}$ ,  $\beta \circ \alpha = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 2 & 5 & 1 & 4 \end{pmatrix}$ ,

$\alpha^2 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 1 & 2 & 4 & 5 \end{pmatrix}$